

Christian-Albrechts-Universität zu Kiel  
 Institut für Informatik  
 Marine Data Science  
 Kiel, Germany



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## Academic Appointments

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|-----------------------------------------------------------------------------------------|--------------------|-------------|
| <b>Christian-Albrechts-Universität zu Kiel</b><br><i>KiTE Research Fellow</i>           | Kiel, Germany      | 2024 – Now  |
| <b>Ukrainian Catholic University</b><br><i>Assistant Professor</i>                      | Lviv, Ukraine      | 2023 – 2024 |
| <b>Czech Inst. of Informatics, Robotics &amp; Cybernetics</b><br><i>Research Fellow</i> | Prague, Czechia    | 2022 – 2023 |
| <b>Chalmers University of Technology</b><br><i>Post-Doctoral Research Scientist</i>     | Gothenburg, Sweden | 2021        |
| <b>Facebook Reality Labs, AR/VR</b><br><i>Post-Doctoral Research Scientist</i>          | Pittsburgh, PA     | 2019 – 2021 |

## Education

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|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
| <b>Czech Technical University</b><br><i>Ph.D., Computer Science</i><br>Thesis: “Methods for the Rectification of Imaged Coplanar Repeated Patterns” | Prague, Czechia | 2020 |
| <b>Czech Technical University</b><br><i>M.Sc., Computer Science</i>                                                                                 | Prague, Czechia | 2013 |
| <b>The University of North Texas</b><br><i>B.Sc., Mathematics</i>                                                                                   | Denton, TX      | 2002 |

## Publications

- J. Pritts**, F. Seegräber, and K. Köser. RANSAC Scoring Done Right. *arXiv preprint arXiv:2606.27385*, 2026
- E. Dexheimer, P. Peluse, J. Chen, **J. Pritts**, and M. Kaess. Information-theoretic online multi-camera extrinsic calibration. *IEEE Robotics and Automation Letters*, 7(2):4757–4764, 2022
- Y. Lochman, K. Liepieshov, J. Chen, M. Perdoch, C. Zach, and **J. Pritts**. Babelcalib: a universal approach to calibrating central cameras. In *ICCV*, 2021
- Y. Lochman, O. Dobosevych, R. Hryniv, and **J. Pritts**. Minimal Solvers for Single-View Auto-Calibration. In *WACV*, 2021
- J. Pritts**, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal solvers for rectifying from radially-distorted conjugate translations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):3931–3948, 2021
- J. Pritts**, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal Solvers for Rectifying from Radially-Distorted Scales and Change of Scales. *International Journal of Computer Vision*, 128(4):950–968, 2020
- J. Pritts**, Z. Kukelova, V. Larsson, and O. Chum. Rectification from Radially-Distorted Scales. In *ACCV*, 2018
- J. Pritts**, Z. Kukelova, V. Larsson, and O. Chum. Radially-Distorted Conjugate Translations. In *CVPR*, 2018
- J. Pritts**, D. Rozumnyi, M. P. Kumar, and O. Chum. Coplanar Repeats by Energy Minimization. In *BMVC*, 2016

**J. Pritts**, O. Chum, and J. Matas. Detection, Rectification, and Segmentation of Coplanar Repeated Patterns. In *CVPR*, 2014

**J. Pritts**, O. Chum, and J. Matas. Approximate Models for Fast and Accurate Epipolar Geometry Estimation. In *IVCNZ*, 2013

## Recognition and Awards

Asian Conference on Computer Vision (ACCV) Saburo Tsuji Best Paper Award 2018  
for “Rectification from Radially-Distorted Scales”

Image and Vision Computing New Zealand (IVCVNZ) Best Paper Award 2013  
for “Approximate Models for Fast and Accurate Epipolar Geometry Estimation”

## Professional Association Memberships

*ELLIS* Scholar 2024-Now

*Kiel Marine Science* Associate 2025-Now

## Funding

*Principal* Kiel Marine Science (KMS) 2025  
“Enhancing Underwater Robotics for Visual Mapping in Turbid Environments”  
Awarded: €10,000

*Principal* Kiel Training for Excellence (KiTE) co-funded by MSCA 2024 – 2027  
“Live and Robust Visual Localization and Mapping in Coastal Waters”  
Awarded: \$225,000

*Contributing* Facebook Sponsored Research Agreement 2020 – 2021  
“In-the-field Extrinsic Calibration of Multi-camera Systems”  
Awarded: \$150,000

*Principal* Facebook Sponsored Research Agreement 2020  
“Calibration of Head-Mounted Multi-Camera Capture Systems”  
Awarded: \$25,000

## Professional Service Activities

*Reviewer* CVPR, ICCV, ECCV, 3DV, WACV, ACCV, ICRA, IJCV, PAMI

*Conference Organizer* Poster Chair 3DV 2022

*Tutorial Organizer* “Affine Correspondences and Their Applications” 3DV 2022

*Tutorial Organizer* “Affine Correspondences and Their Applications” CVPR 2022

## Teaching

3D Computer Vision 2025 – 2026  
Substitute Instructor, M.Sc. level, Kiel University

Image Retrieval 2017 – 2018  
Instructor, M.Sc. level, Ukrainian Catholic University

Pattern Recognition and Machine Learning 2013 – 2016  
Teaching Assistant, B.Sc. level, Czech Technical University in Prague

## Advising

### M.Sc. Students

|                   |                                                                                                                                     |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Hendrik Sauer     | Kiel University                                                                                                                     | in progress |
|                   | Thesis: “Affine-Equivariant Neural Networks for the Feature Matching Problem”                                                       |             |
| Ties Harder       | Kiel University                                                                                                                     | in progress |
|                   | Thesis: “Boundary Movement Aware Differentiable Rendering for Visual Calibration Tasks”                                             |             |
| Philipp Böll      | Kiel University                                                                                                                     | in progress |
|                   | Thesis: “Underwater Light Rig Auto-Calibration from Asynchronous Cameras”                                                           |             |
| Till Sittart      | Kiel University                                                                                                                     | 2026        |
|                   | Thesis: “Affine-Equivariant Blob Targets for Robust Absolute Pose in Robotics”                                                      |             |
| Liam Boddin       | Kiel University                                                                                                                     | 2025        |
|                   | Thesis: “Underwater Calibration of Near-point Anisotropic Lights”                                                                   |             |
| Igor Babin        | Ukrainian Catholic University                                                                                                       | 2024        |
|                   | Thesis: “Image Inpainting in Latent Space”                                                                                          |             |
| Dmytro Nadobko    | Ukrainian Catholic University                                                                                                       | 2023        |
|                   | Thesis: “Supervised Learning of Correspondence Volumes for Coplanar Repetitive Patterns”                                            |             |
| Andrii Stadnik    | Ukrainian Catholic University                                                                                                       | 2023        |
|                   | Thesis: “Corner Localization and Camera Calibration from Imaged Lattices”                                                           |             |
| Yaroslava Lochman | Ukrainian Catholic University                                                                                                       | 2018 – 2020 |
|                   | Thesis: “Minimal Solvers for Single-View Auto-Calibration”<br><i>moved on as Ph.D. student at Chalmers University of Technology</i> |             |

### B.Sc. Students

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Yaroslav Romanus      | Ukrainian Catholic University                                                                                                                                                             | 2024        |
|                       | Thesis: “Applying Motion in Latent Space”                                                                                                                                                 |             |
| Ostap Viniavskyi      | Ukrainian Catholic University                                                                                                                                                             | 2019 – 2021 |
|                       | Thesis: “Learning Discriminative Context-Aware Keypoints Representations for Resolving Ambiguous Matches”<br><i>moved on as Researcher at The ML Lab at Ukrainian Catholic University</i> |             |
| Kostiantyn Liepieshov | Ukrainian Catholic University                                                                                                                                                             | 2019 – 2021 |
|                       | Thesis: “Manhattan-Frame Detection in Lens-Distorted Images”<br><i>moved on as M.Sc. student at Ukrainian Catholic University</i>                                                         |             |

## Invited Talks and Lectures

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| 3DV Tutorial Talk<br>“Just One Image is All It Takes...”                                                                                  | September 12, 2022<br>Prague, Czechia |
| CVPR Tutorial Talk<br>“Just One Image is All It Takes...”                                                                                 | June 19, 2022<br>New Orleans, LA      |
| Facebook Reality Labs<br>“Minimal Solvers and Multi-Model Estimators for Image Undistortion and Scene-Plane Rectification”                | June 14, 2018<br>Pittsburgh, PA       |
| The Aspen Institute’s 2018 Young Leader’s Program<br>“Opportunities and Risks of Artificial Intelligence”                                 | March 17, 2018<br>Tále, Slovakia      |
| The Eastern European Computer Vision Conference<br>“Detection, Rectification, and Segmentation of Coplanar Repeated Patterns”             | July 14, 2017<br>Odesa, Ukraine       |
| The 34th Pattern Recognition and Computer Vision Colloquium<br>“Detection, Rectification, and Segmentation of Coplanar Repeated Patterns” | April 3, 2014<br>Prague, Czechia      |